$$I_n = n(n-1)(n-2)\cdots(2)(1)I_0 .$$

But, $I_0 = \int_0^\infty e^{-x}x^0dx = \left(-e^{-x}\right)_0^\infty = 0+1=1$. So, we get $I_n = n(n-1)(n-2)\cdots(2)(1) = n!$.

Hence, we get

Result

$\int_0^\infty e^{-x}x^n dx = n!$, where n is a nonnegative integer.

Note

The integral $\int_0^\infty e^{-x}x^{n-1}dx$ defines a unique positive integer for every positive integer $n \geq 1$.

Definition 9.1

$\int_0^\infty e^{-x}x^{n-1}dx$ is called the **gamma integral**. It is denoted by $\Gamma(n)$ and is read as "gamma of n".

Note

$$\Gamma(n+1) = n\Gamma(n) .$$

$$\Gamma(1) = \int_0^\infty e^{-x}x^0dx = \left(-e^{-x}\right)_0^\infty = 0+1=1 ,$$

$$\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}dx .$$

$$= (n-1)!, n = 1, 2, 3, \ldots$$

Example 9.44

Evaluate $\int_0^\infty e^{-ax}x^n dx$,where $a > 0$.

Solution

Making the substitution $t = ax$, we get $dt = adx$ and $x = 0 \Rightarrow t = 0$ and $x = \infty \Rightarrow t = \infty$.

Hence, we get

$$\int_0^\infty e^{-ax}x^n dx = \int_0^\infty e^{-t}\left(\frac{t}{a}\right)^n \frac{dt}{a} = \frac{1}{a^{n+1}}\int_0^\infty e^{-t}t^n dt$$

$$= \frac{1}{a^{n+1}}\int_0^\infty e^{-x}x^n dx = \frac{n!}{a^{n+1}} .$$

Thus

$$\int_0^\infty e^{-ax}x^n dx = \frac{n!}{a^{n+1}}$$

Example 9.45

Show that $\Gamma(n) = 2\int_0^\infty e^{-x^2}x^{2n-1}dx.$

Solution

Using the substitution $x = \sqrt{u}$, we get $dx = \frac{1}{2\sqrt{u}}du$.

When $x = 0$, we get $u = 0$ .When $x = \infty$, we get $u = \infty$.

$$\therefore 2\int_0^\infty e^{-x^2}x^{2n-1}dx = 2\int_0^\infty e^{-u}\left(\sqrt{u}\right)^{2n-1}\frac{1}{2\sqrt{u}}du = \int_0^\infty e^{-u}u^{n-1}du = \Gamma(n) .$$

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